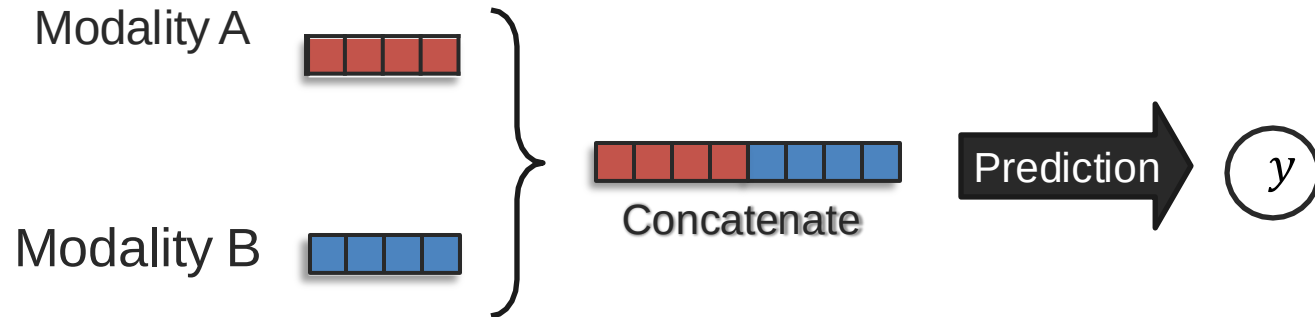
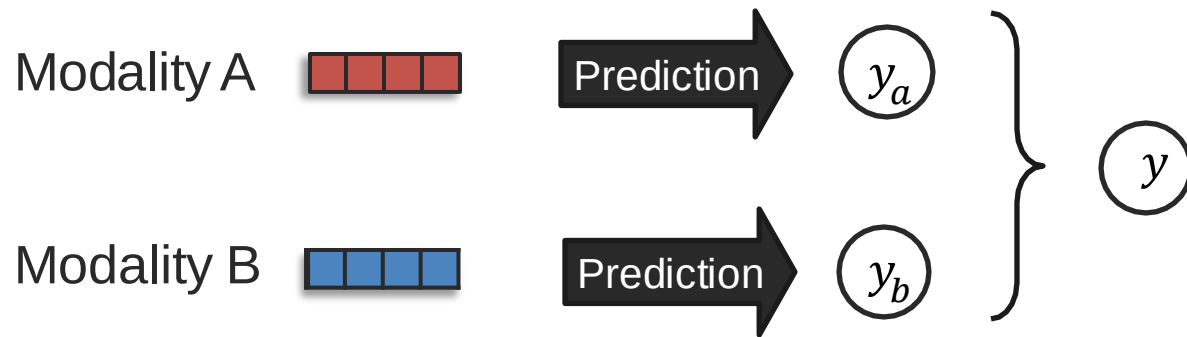


Early and Late Fusion – A historical view

Early fusion:



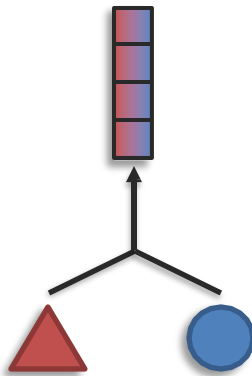
Late fusion (ensemble):



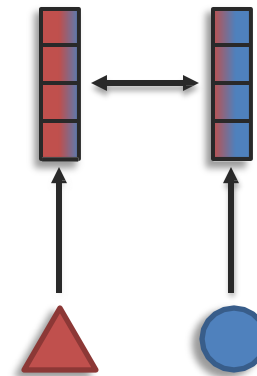
Challenge 1: Representation

Definition: Learning representations that reflect cross-modal interactions between individual elements, across different modalities

Fusion

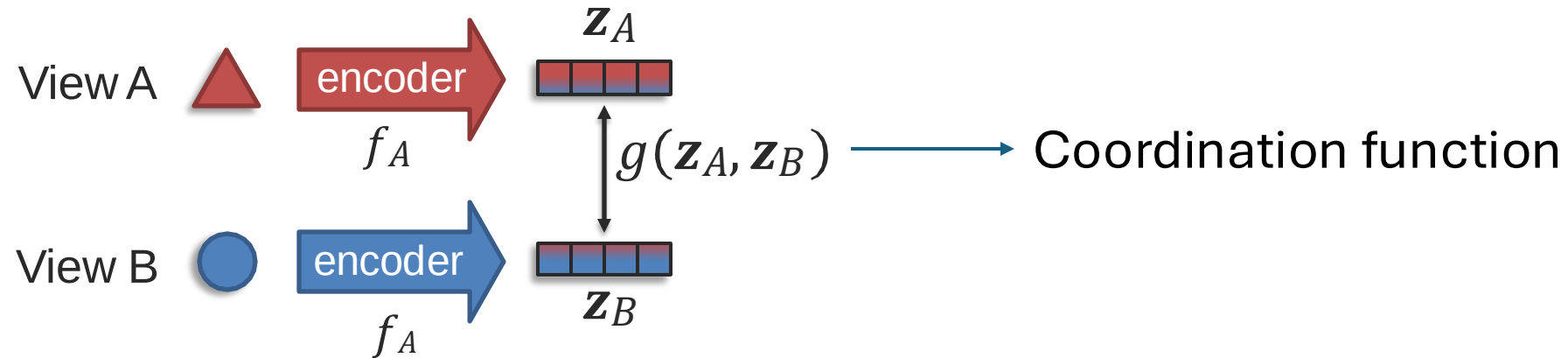


Coordination



Today

Coordination Function



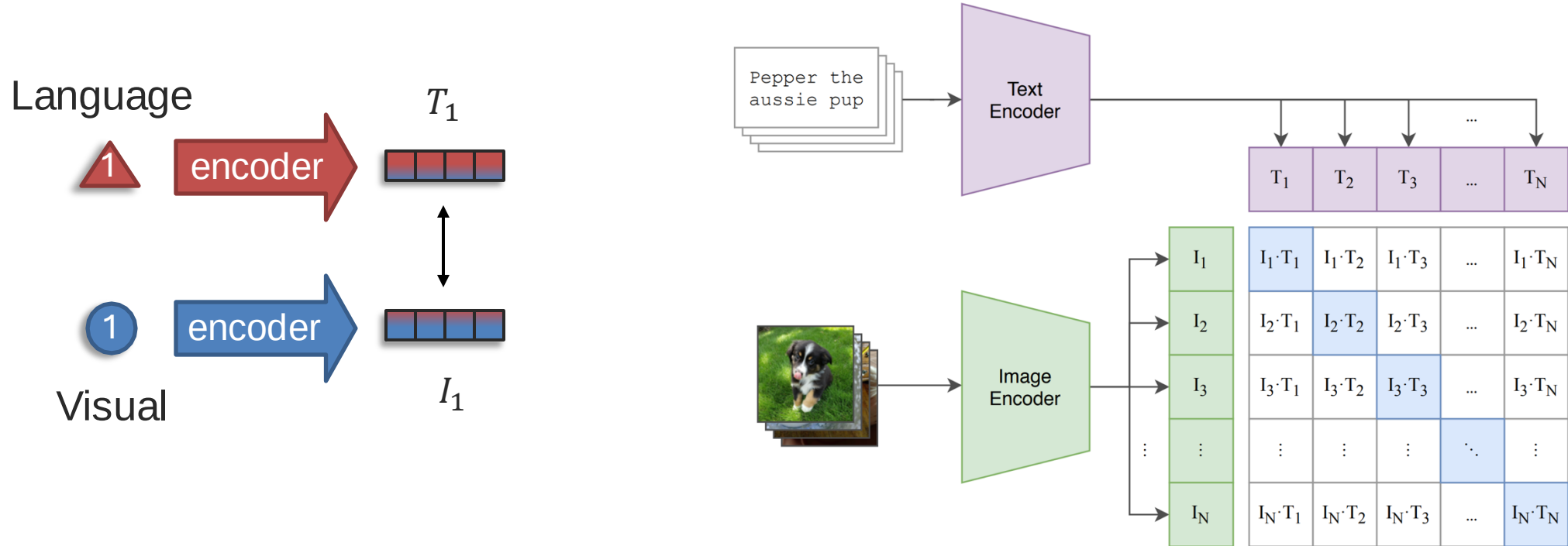
Cosine similarity:

$$g(z_A, z_B) = \frac{z_A \cdot z_B}{\|z_A\| \|z_B\|}$$

Strong coordination!

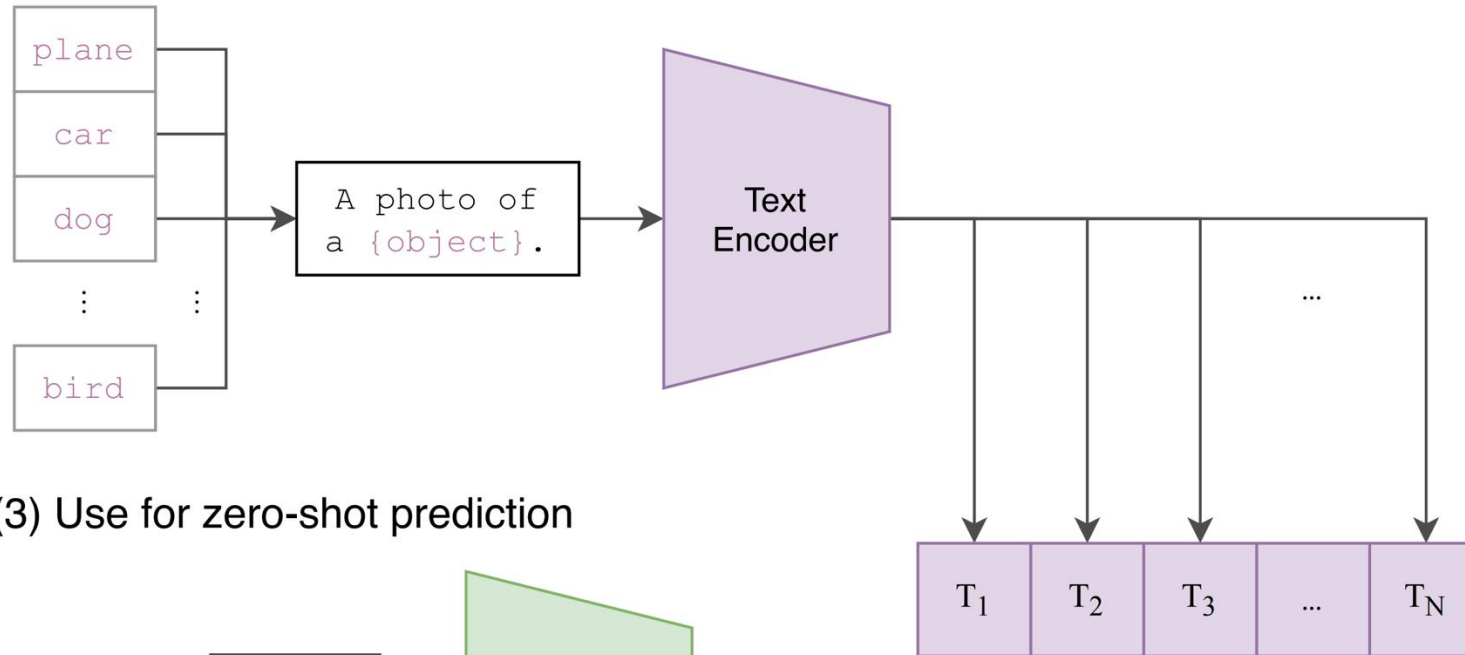
Contrastive Language–Image Pre-training

- Dataset of 400 million (image, text) pairs collected from the internet

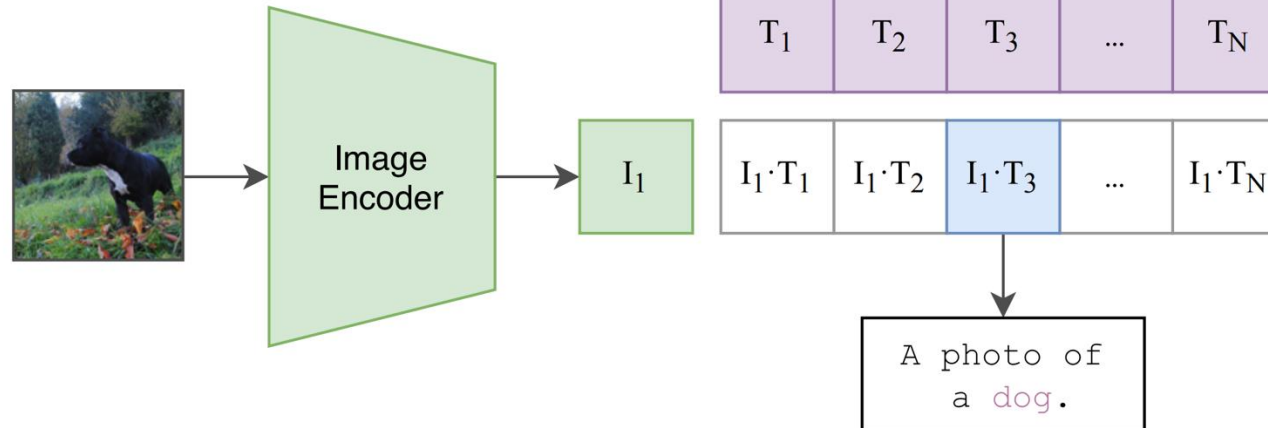


Zero-shot classification

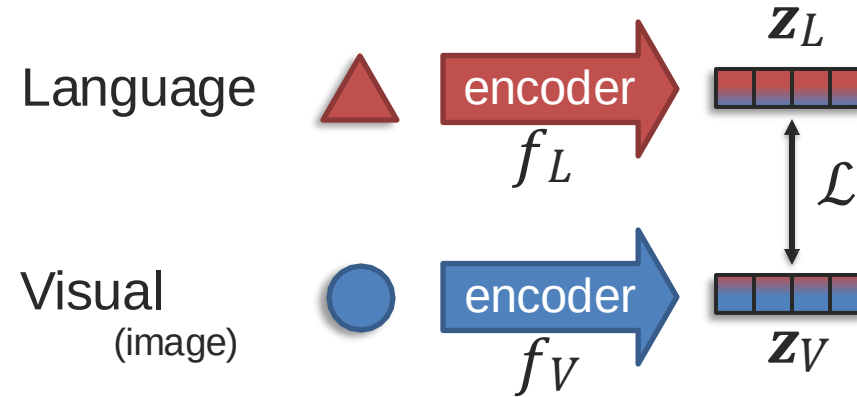
(2) Create dataset classifier from label text



(3) Use for zero-shot prediction



CLIP's contrastive loss



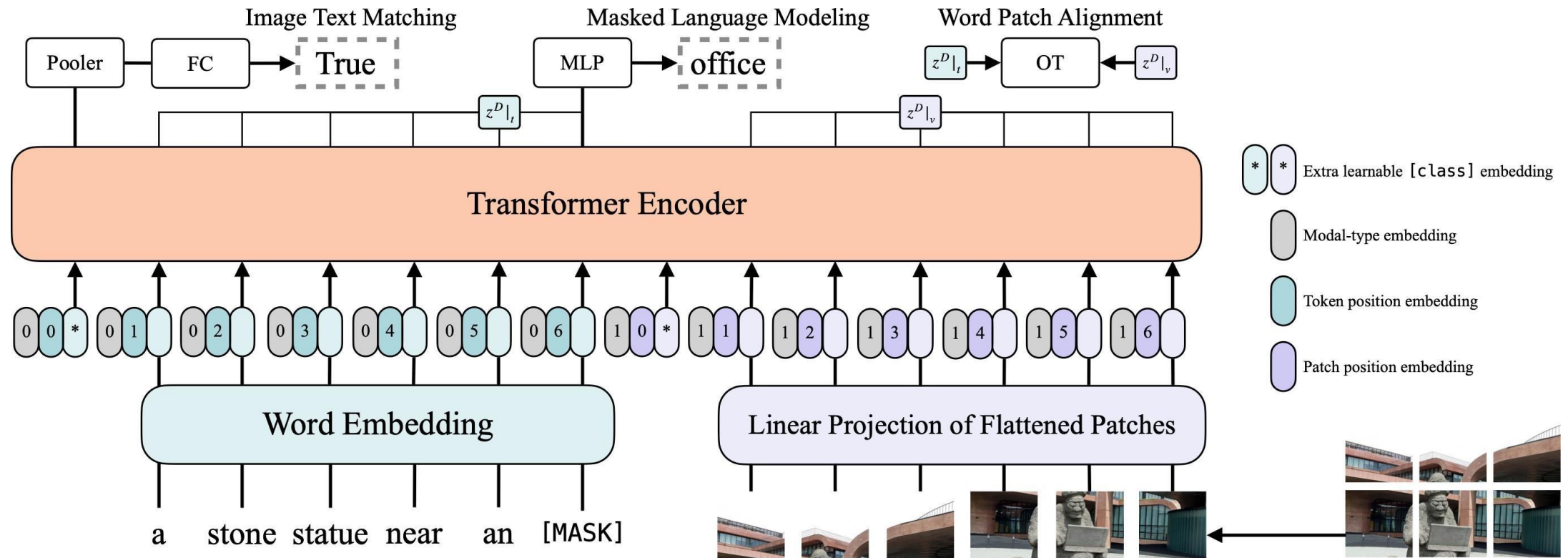
Popular contrastive loss: InfoNCE

$$\mathcal{L} = -\frac{1}{N} \sum_{i=1}^N \log \frac{\text{sim}(\mathbf{z}_A^i, \mathbf{z}_B^i)}{\sum_{j=1}^N \text{sim}(\mathbf{z}_A^i, \mathbf{z}_B^j)}$$

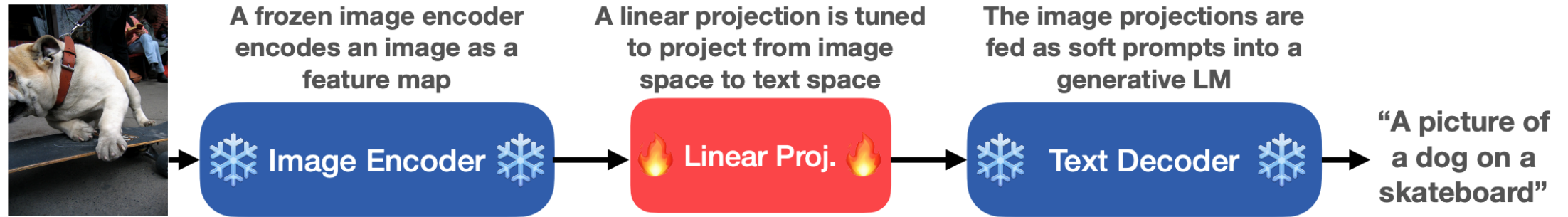
Annotations for the formula:

- A green arrow points from the text 'positive pairs' to the numerator $\text{sim}(\mathbf{z}_A^i, \mathbf{z}_B^i)$.
- An orange arrow points from the text 'Similarity function can be cosine similarity' to the sim function in the denominator.
- A red arrow points from the text 'negative pairs and positive pairs' to the denominator $\sum_{j=1}^N \text{sim}(\mathbf{z}_A^i, \mathbf{z}_B^j)$.

Visual-and-Language Transformer (ViLT) ($\approx$ BERT + ViT)

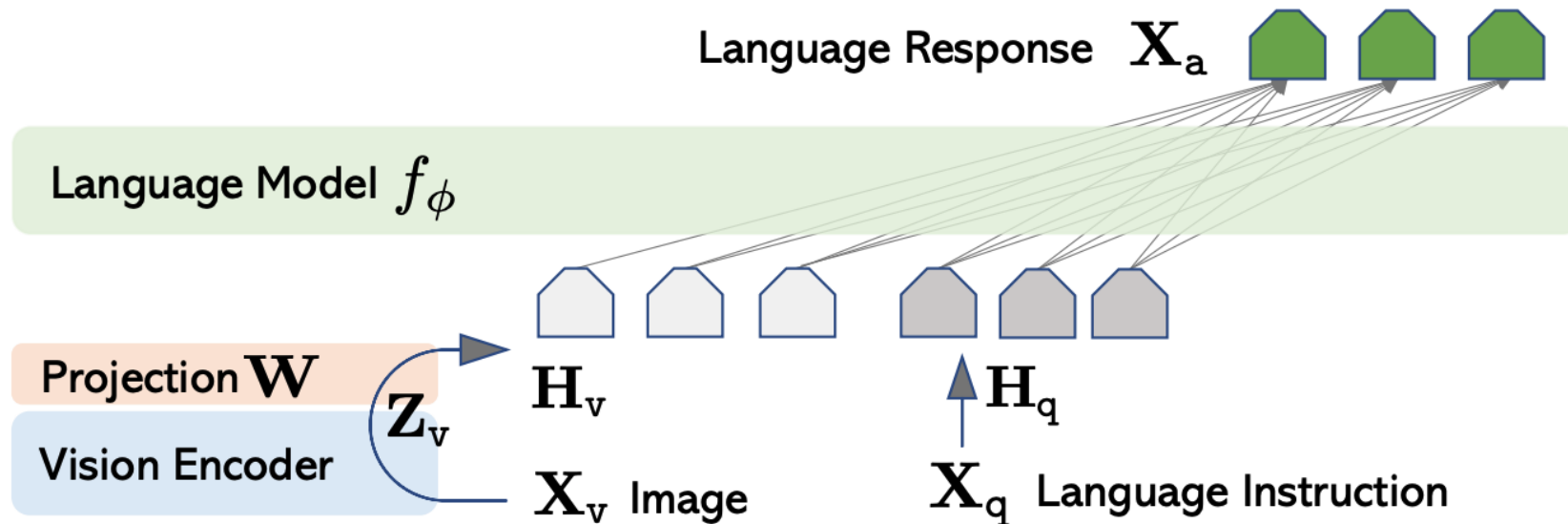


Linear mapping from image to text

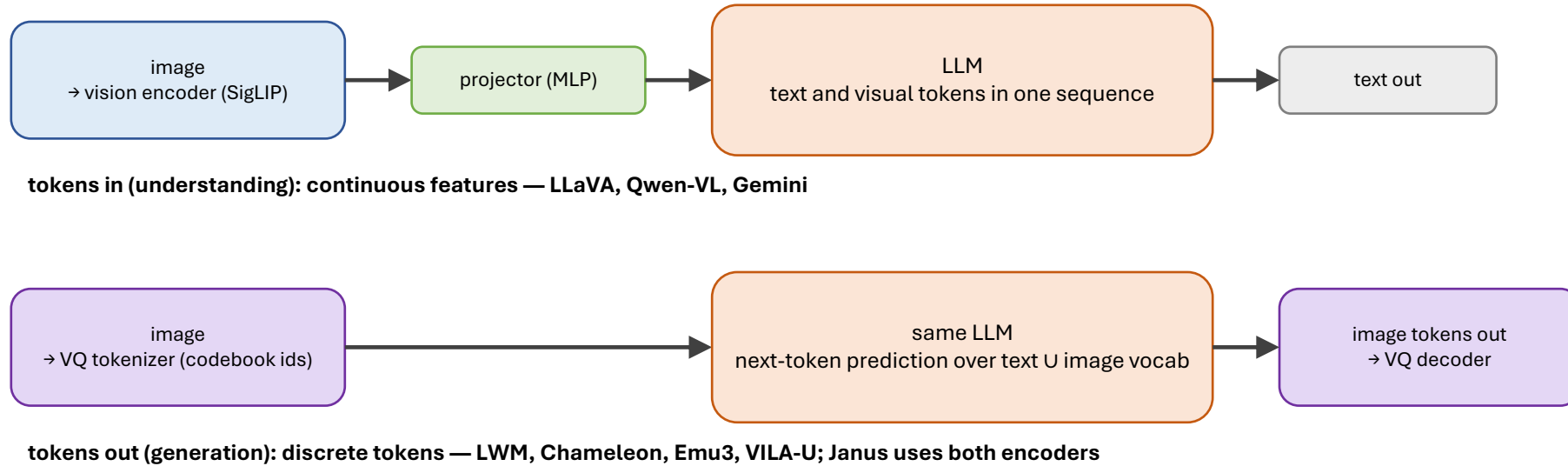


Visual Instruction Tuning

- Conditioning with simple linear adaptars



The recipe has not changed; the details have



Understanding: vision encoder (CLIP → SigLIP 2) → projector → LLM; visual features enter as continuous tokens (Lecture 9)
2025–26 changes: **native resolution** (variable number of patches, 2D RoPE), video frames with **timestamps** in the text stream, multi-layer visual features, three-stage training (align → multi-task → SFT)
Generation route (bottom): tokenize the image with a VQ codebook and let the *same* LLM emit image ids

Next-token image generation grew up: Chameleon, Emu3, VAR

- **Chameleon** (Meta, May 2024): early fusion — text and VQ image tokens in one vocabulary, one transformer trained from scratch on both
- **Emu3** (BAAI, Sep 2024): next-token prediction only, for images, video and text — competitive with SDXL and LLaVA-1.6 in the authors' evaluation
- **VAR** (Tian et al., NeurIPS 2024 best paper): predict the image coarse-to-fine, **one scale at a time** instead of raster order; GPT-style AR beat diffusion transformers on ImageNet FID
- Lesson: the tokenizer and the prediction order matter as much as the transformer

Janus: decouple the two visual encoders, keep one transformer

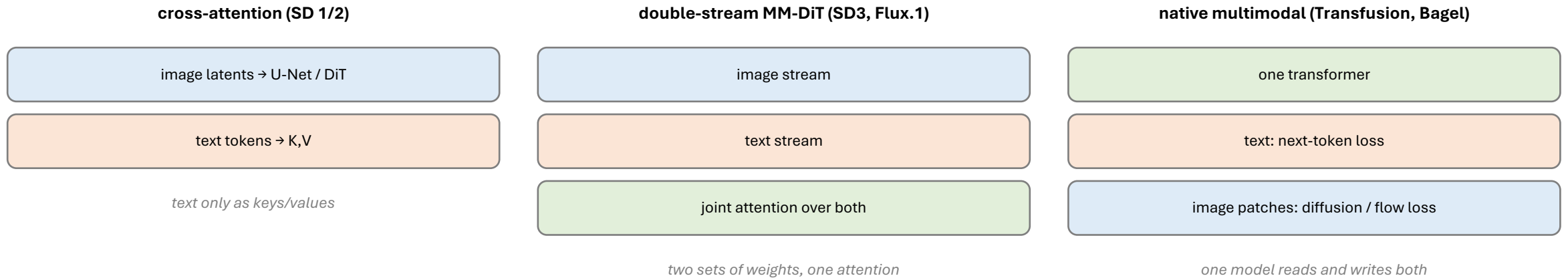


decoupling removes the conflict between semantic features (understanding) and pixel detail (generation)

Understanding and generation need different information (semantics vs pixel detail); one encoder compromises both
Janus (DeepSeek, Oct 2024): a SigLIP-style encoder for understanding, a VQ tokenizer for generation, one autoregressive transformer that also emits image tokens; the authors report it surpasses previous unified models and matches task-specific ones

Janus-Pro (Jan 2025): scaled data and training, 7B; **JanusFlow** (Nov 2024): the generation branch becomes rectified flow (Lecture 10b)

Mixing objectives in one transformer

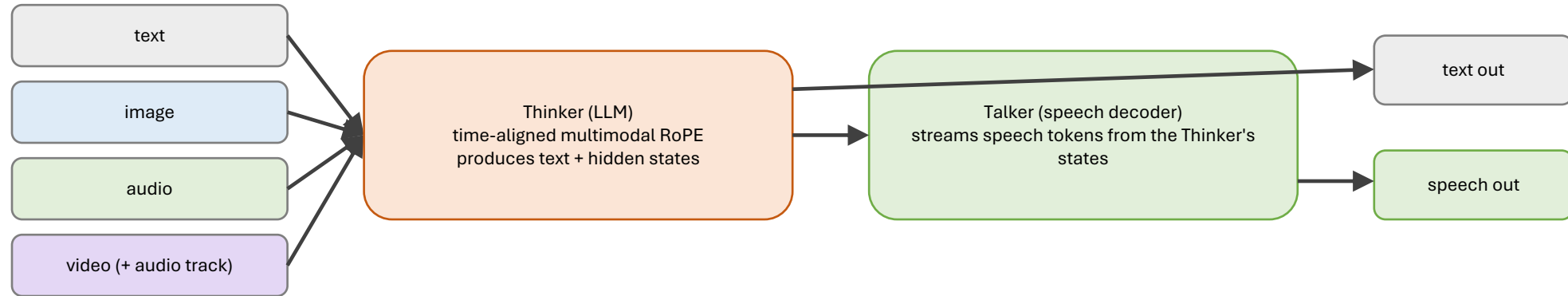


Transfusion (Meta, Aug 2024): one transformer; next-token loss on text tokens, diffusion loss on continuous image patches; markers delimit image spans

Show-o (Aug 2024): AR text + *discrete* diffusion over image tokens; **Bagel** (ByteDance, May 2025): mixture-of-transformer-experts, 7B active, trained on interleaved video/web data — understanding, generation and editing

Products: **GPT-4o native image generation** (Mar 2025), Gemini's **Nano Banana** (Aug 2025) and Nano Banana Pro (Nov 2025) — editing with the chat model's world knowledge and multi-turn context; internals not public

Any-to-any in 2025: Thinker–Talker



Qwen2.5-Omni (Mar 2025) and Qwen3-Omni (Sep 2025): any-to-any with a two-part design; GPT-4o (2024) and Gemini Live use single end-to-end models (details not public)

Qwen2.5-Omni (Mar 2025): a Thinker LLM consumes text, image, audio and video with time-aligned multimodal RoPE; a Talker streams speech from the Thinker's hidden states

Qwen3-Omni (Sep 2025): 119 languages, long audio input, real-time speech output, a thinking mode

CS231n's closing note (May 2026): omni models started with GPT-4o (2024); new omni models from Thinking Machines and Google in May 2026